

Research Note 01

Seasonal Structure and Residual Modeling of California Highway Traffic Flow

Traffic Forecasting Project

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Abstract

We study one-week-ahead ($H=168$ hourly) forecasting of freeway traffic *flow* in the Greater Bay Area, using five years of Caltrans loop-detector data (LargeST). We show that a trivial *seasonal baseline* – the average of the same hour-of-week over the previous four weeks – explains roughly 94% of the variance of hourly flow, per cell (sensor $\times$ hour). We then study the *seasonal residual* $r = y - b$, establish that for a one-week-ahead task only *weekly* lags are usable, and evaluate a seasonal autoregression on the residual. It yields a small but consistent out-of-sample gain ($\sim 3\%$ RMSE), and its fitted coefficients reveal that the real signal is a *re-weighting* of the flat four-week average toward more recent weeks. We close with concrete next steps.

Key findings

- Traffic **volume** is dominated by a strong, stable weekly cycle (commute peaks; weekday/weekend contrast).
- A per-cell **4-week hour-of-week mean baseline** explains $> 90\%$ of variance: pooled $R^2 = 0.936$, per-sensor median $R^2 = 0.913$, stable across 2017–2019.
- The **seasonal residual** has strong *short-lag* autocorrelation (AR(1)-like, lag-1 = 0.81) but only weak/negative *weekly-lag* autocorrelation – the latter partly an artifact of the baseline filter.
- For **week-ahead** forecasting, short lags are unusable; only weekly lags are available across the whole horizon. A seasonal-residual AR on weekly lags gives $\sim 3\%$ RMSE improvement and, interpreted as weights on raw weekly values, **learns a decaying (EWMA-like) weekly average**.

1 The dataset

Source. The data come from the Caltrans Performance Measurement System (PeMS), which aggregates measurements from inductive-loop detectors embedded in the freeway pavement across California. Each *station* (which we also call a *sensor* or *spatial unit*) reports, in five-minute bins, three quantities: *flow* (vehicle count per unit time), *occupancy* (fraction of time the detector is covered by a vehicle), and *speed*. We forecast **flow**, the most directly seasonal of the three; speed is far noisier (dominated by incident-driven congestion) and, in a parallel check on the PeMS-BAY speed dataset, the same baseline reached only $R^2 \approx 0.76$ versus 0.94 for flow.

Benchmark and region. We use **LargeST** (Liu et al., NeurIPS 2023), which assembles 8,600 PeMS sensors statewide at 5-minute resolution for 2017–2021. We restrict to **Caltrans District 4 (Greater Bay Area)**, $N = 2,352$ sensors, and – to avoid the COVID-19 disruption – use the pre-pandemic years **2017–2019** as the working set. (2020 is cached separately for a future stress test; LargeST ends in 2021, so genuinely post-pandemic data would have to be pulled from the PeMS Clearinghouse.)

Aggregation and notation. We average the 5-minute series to hourly (12 bins $\rightarrow$ 1 hour). The result is a matrix

$$Y \in \mathbb{R}^{T \times N}, \quad y_{i,t} = \text{mean flow (veh/h) at sensor } i \text{ during hour } t,$$

with $T = 26,280$ hourly steps (3 years) and $N = 2,352$ sensors. Each sensor i is a fixed location, freeway, and direction (e.g. I-80 W). A typical sensor carries a median flow of ~ 220 veh/h. Missing values (NaN, $\sim 2\%$ of hours) are imputed per sensor with that sensor’s hour-of-week mean. We define the weekly period

$$P = 24 \times 7 = 168 \text{ hours},$$

and the *hour-of-week* index $\phi(t) = t \bmod P \in \{0, \dots, 167\}$, which encodes both the day-of-week and the hour-of-day of timestamp t .

2 Exploratory analysis: a strong weekly cycle

Figure 1 shows the mean flow as a function of hour-of-week for three sensors (busy / median / quiet). The structure is striking and highly regular: each weekday shows the classic *double peak* (morning and evening commutes); weekends collapse into a single broad midday hump; and the shape repeats week after week. Figure 2 renders the same information for one busy sensor as a day-of-week $\times$ hour-of-day heatmap.

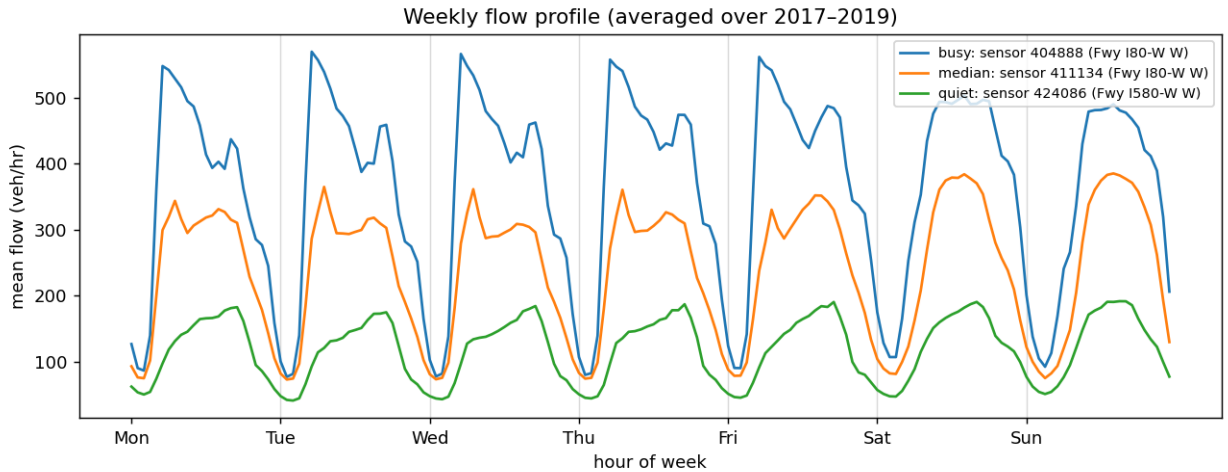


Figure 1: Mean flow by hour-of-week (averaged over 2017–2019) for three sensors. Vertical lines mark day boundaries. Weekday commute double-peaks and the weekend single-hump are visible and stable across all seven days.

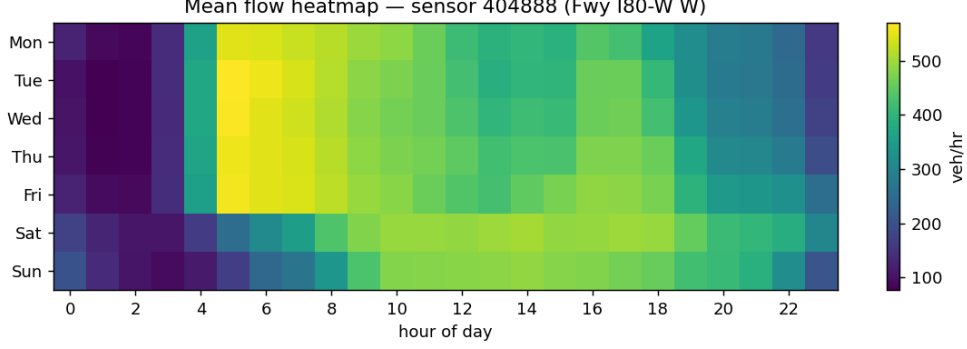


Figure 2: Mean flow heatmap (weekday $\times$ hour-of-day) for a busy I-80 W sensor. The bright morning/evening commute bands on weekdays and the diffuse weekend pattern summarize the dominant seasonal signal.

3 The seasonal baseline

Because the weekly cycle is so dominant, a natural predictor for any cell is the recent average at the *same hour of week*. We define the four-week **seasonal baseline**

$$b_{i,t} = \frac{1}{4} \sum_{k=1}^4 y_{i,t-Pk}, \quad P = 168. \quad (1)$$

Every lag is a multiple of one week, so $b_{i,t}$ averages the four most recent realizations of the same hour-of-week.

It is a valid one-week-ahead forecast. At a forecast origin t , consider a target $s = t + h$ with horizon $h \in \{1, \dots, 168\}$. Every term in $b_{i,s} = \frac{1}{4} \sum_{k=1}^4 y_{i,s-Pk}$ has index $s - Pk \leq t$ for all $k \geq 1$ (since $h \leq P$). Hence the baseline for the *entire* week ahead is known at t with **no leakage and no recursion**.

It explains $> 90\%$ of variance. Define the per-sensor variance explained

$$\text{VE}_i = 1 - \frac{\text{Var}_t(r_{i,t})}{\text{Var}_t(y_{i,t})}, \quad r_{i,t} = y_{i,t} - b_{i,t}.$$

Pooled over all cells, $\text{VE} = 0.936$; the per-sensor distribution (Figure 3) has median 0.913 and 10th–90th percentiles $[0.845, 0.950]$. In a full rolling-origin backtest (origins stepped daily, $H = 168, 1,061$ forecasts), the baseline attains the metrics in Table 1, stable year to year ($R^2 = 0.941/0.928/0.940$ for 2017/2018/2019). Overall the baseline reduces the standard deviation of flow from 164.6 veh/h to a residual 41.7 veh/h – it removes about three-quarters of the typical fluctuation.

Table 1: Rolling-origin one-week-ahead backtest, GBA 2017–2019 (1,061 weekly forecasts $\times$ 2,352 sensors).

Forecaster	Pooled R^2	RMSE (veh/h)	Per-sensor median R^2
Hour-of-week mean, 4 weeks	0.936	41.7	0.913
Hour-of-week mean, 8 weeks	0.937	41.2	0.915
Same hour, last week only	0.910	49.5	0.878

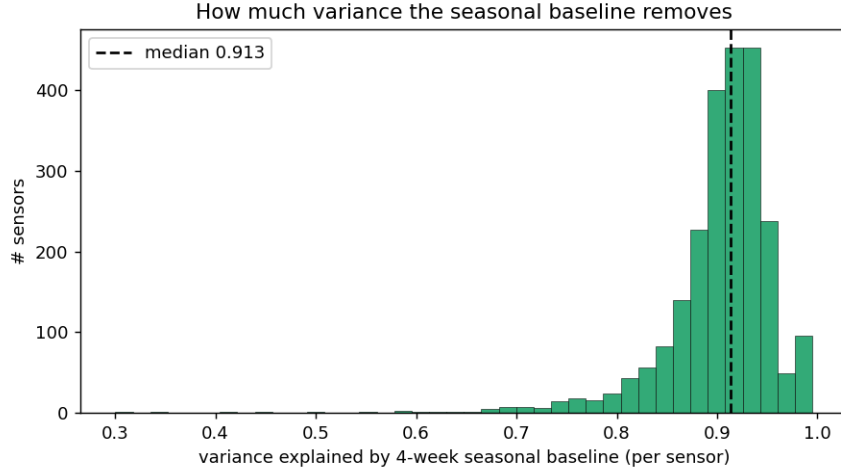


Figure 3: Distribution over sensors of variance explained by the 4-week seasonal baseline. Even the hardest cells exceed 0.84; there is no long unpredictable tail (unlike speed).

4 The seasonal residual

We now model what the baseline leaves behind, the **seasonal residual**

$$r_{i,t} = y_{i,t} - b_{i,t} = \left(1 - \frac{1}{4} \sum_{k=1}^4 B^{Pk}\right) y_{i,t}, \quad (2)$$

where B is the backshift operator ($B^m y_t = y_{t-m}$). Equation (2) makes explicit that the baseline is a *seasonal filter*: r is y with a weekly moving-average level removed. Figure 4 shows the decomposition for one sensor over three weeks; the baseline tracks the shape tightly and the residual is the harder, irregular remainder.

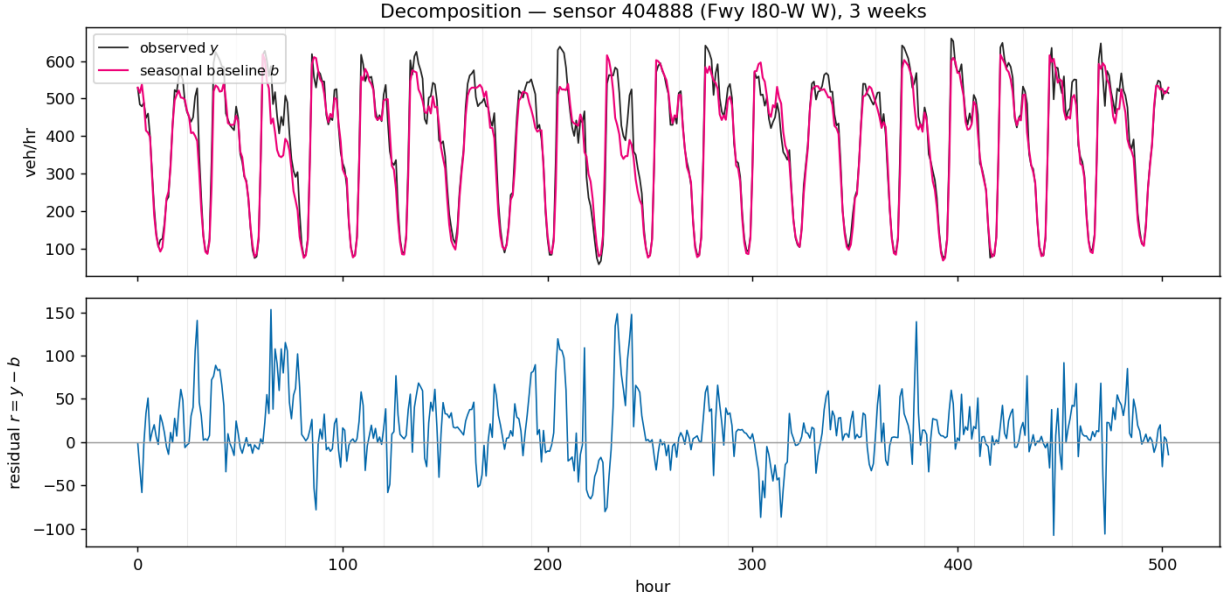


Figure 4: Top: observed flow y and seasonal baseline b for a busy sensor over three weeks. Bottom: the residual $r = y - b$.

4.1 Autocorrelation structure

Figure 5 and Table 2 summarize the mean residual autocorrelation (over 250 sensors). Three observations:

1. **Short lags dominate and are nearly pure AR(1).** The ACF decays as 0.806^k for the first few hours (0.806, 0.650, 0.525) and the partial ACF cuts off after lag 1 (+0.806, then ≈ 0). There is a secondary *daily* echo (lag 24 = 0.331).
2. **The weekly lag is weak** (+0.094 at lag 168).
3. **The 2–4 week lags are negative** (−0.05, −0.15, −0.18). This is *not* physical: by construction (2) subtracts the previous four same-hour-of-week values, so r_t is mechanically anti-correlated with those weeks. The negative weekly tail is the fingerprint of the baseline filter, not week-over-week signal.

Table 2: Mean residual autocorrelation at selected lags (250 sensors).

Lag (h)	1	2	3	24	48	168	336	504/672
ACF	+0.806	+0.650	+0.525	+0.331	+0.215	+0.094	−0.049	−0.149/−0.181

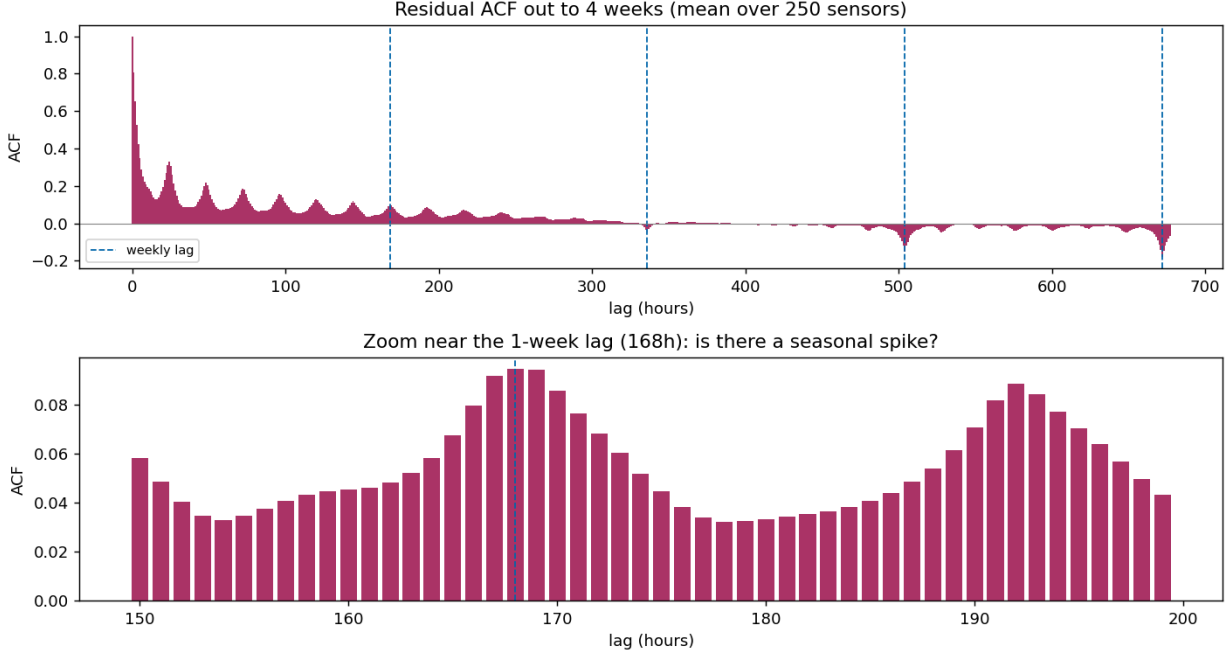


Figure 5: Residual ACF out to four weeks (top) and zoomed near the one-week lag (bottom). Dashed lines mark weekly lags. Note the daily echoes in the short-lag region, the modest bump at 168h, and the growing *negative* values at 2–4 weeks induced by the baseline filter.

5 Week-ahead constraint and the seasonal-residual AR

Why short lags are useless here. The lag-1 autocorrelation of 0.81 is irrelevant to a one-week-ahead forecast. At origin t , predicting target $s = t + h$, the freshest usable observation is at t , i.e. h hours before the target; at the far end of the horizon ($h = 168$) that is a *full week* before s . An hourly AR(1) iterated forward decays as $0.81^h \rightarrow 0$ over the week and contributes essentially nothing. **The only lags available across the entire horizon are the weekly ones.**

Model. We therefore model the seasonal residual from its own weekly lags – a seasonal autoregression:

$$\hat{r}_{i,s} = \psi_0 + \sum_{m=1}^M \psi_m r_{i,s-Pm}, \quad \hat{y}_{i,s} = b_{i,s} + \hat{r}_{i,s}. \quad (3)$$

Crucially, each feature $r_{i,s-Pm}$ depends only on y up to time $s - P \leq t$, so it is observed at the origin for *every* horizon $h \in \{1, \dots, 168\}$. The model needs no recursion and applies uniformly across the horizon.

Out-of-sample result. We fit (3) by pooled OLS on 2017–2018 and test on 2019 (Table 3). The seasonal-residual AR gives a small but consistent improvement ($\sim 3\%$ RMSE; residual $R^2 \approx 0.05$). Two sharpenings of the picture:

- **A single weekly lag is essentially useless** (residual $R^2 = 0.004$). All the value comes from *combining* weeks.

- The fitted coefficients are negative and growing,

$$\psi_1 = +0.04, \quad \psi_2 = -0.07, \quad \psi_3 = -0.14, \quad \psi_4 = -0.20,$$

echoing the negative ACF tail.

Table 3: Seasonal-residual AR, trained 2017–18, tested 2019 (pooled OLS, 600-sensor subsample). Baseline figures here differ slightly from Table 1 due to the single-split, subsampled setup.

Model	Residual R^2	One-week-ahead y R^2	RMSE (veh/h)
Baseline only ($\hat{r} = 0$)	—	0.9417	39.65
+ seasonal-AR, 4 weekly lags	0.051	0.9447	38.62 (−2.6%)
+ seasonal-AR, 8 weekly lags	0.058	0.9451	38.48 (−3.0%)
1 weekly lag only	0.004	—	39.56 (−0.2%)

Interpretation: the model is re-weighting the baseline. Substituting $r_{i,s-Pm} = y_{i,s-Pm} - b_{i,s-Pm}$ into (3), the coefficient on the raw weekly value $y_{i,s-Pm}$ becomes $\frac{1}{4} + \psi_m$ for $m = 1, \dots, 4$ (plus smaller corrections from weeks 5–8 carried by the $b_{i,s-Pm}$ terms). With the fitted ψ this gives effective weekly weights of approximately

$$(0.29, 0.18, 0.11, 0.05) \quad \text{versus the flat} \quad (0.25, 0.25, 0.25, 0.25).$$

In other words, the seasonal-residual AR is not exploiting “last week’s anomaly persists” – that signal is near zero. It is discovering that **recent weeks deserve more weight than older ones**, i.e. it converts the flat four-week mean into a *decaying*, *EWMA-like* weekly average.

6 Limitations

- The seasonal-AR result uses a single train/test split and a 600-sensor subsample with *pooled* (global) coefficients; it should be folded into the full rolling-origin harness and refit per sensor.
- Missing hours ($\sim 2\%$) are imputed by hour-of-week means, which can flatter the baseline slightly.
- Flow appears clipped at 999 veh/h in the source; only the extreme tail is affected.
- Results are GBA-only and pre-COVID; robustness across regions and regimes (2020, post-2022) is untested.

7 What to try next

Ordered roughly by expected value-for-effort.

1. **Replace the flat baseline with a learned decaying weekly weight (EWMA).** Section 5 shows the residual model is really re-weighting the baseline. Fitting a single decay rate λ (or free weights over $L > 4$ weeks) directly, $b_{i,t} = \sum_{k \geq 1} w_k y_{i,t-Pk}$ with $\sum_k w_k = 1$, should capture the same gain more parsimoniously and leave a *cleaner* residual for the models below. This is the most direct follow-up.

2. **Calendar / holiday effects.** The baseline mishandles holidays (a holiday target week is averaged against normal weeks, and vice versa). Adding holiday and special-day handling (separate profiles, or an indicator feature) should remove a structured chunk of residual error.
3. **Exogenous covariates, esp. weather.** Rain and major events plausibly drive a portion of the residual and are themselves forecastable a week out (weather forecasts), making them legitimately usable features.
4. **Spatial / global models on weekly-lagged residuals.** Residuals are spatially correlated along corridors. A single *global* model (gradient boosting or a neural net / GNN) over all sensors, with features = several weekly-lagged residuals + calendar + sensor embedding, can share strength across cells. Respect the week-ahead constraint: only weekly-lagged information is available across the full horizon.
5. **Proper evaluation upgrades.** Move the seasonal-AR into the rolling harness; report a *per-horizon* error curve (the seasonal-AR should help uniformly across h since it uses no recursion); add prediction intervals from the residual distribution.
6. **Estimate the predictability ceiling.** Use cross-sensor consistency to estimate how much residual variance is irreducible (incident noise) vs. learnable, so we know how much head-room remains above $R^2 \approx 0.94$.
7. **Regime robustness.** Run the 2020 COVID stress test and, once pulled, post-2022 PeMS data, to see where the seasonal structure breaks.

Reproducibility. Figures and numbers are produced by `src/eda.py`, `src/residual_acf.py`, `src/run_gba.py`, and `src/seasonal_ar.py` in the project repository.